1. Introduction

1.1 Project Overview

This project extends the Eclipse BaSyx AAS Web UI by introducing additional functionalities for processing technical XML-based engineering data, such as KBL, VEC, and other structured industrial formats.

The main objective of these extensions is to simplify and standardize the integration of external engineering data into the Asset Administration Shell (AAS). The system enables users to upload complex model files, automatically validate their structure and content, extract relevant technical information, and visualize XML data in a clear and navigable format.

By automating these processes, the plugins reduce manual effort, improve data consistency, and support a more efficient and reliable engineering workflow within the BaSyx ecosystem.

1.2 Purpose of this Manual

This User Manual explains how to use the implemented functionalities step by step.

It is intended for:

- Users of the BaSyx Web UI
- Developers and testers working with the extensions
- Project partners and stakeholders evaluating the solution

The manual focuses on practical usage rather than technical implementation details and provides a guided overview of all available features.

1.3 Included Functionalities

The project introduces the following extensions to the BaSyx Web UI:

- File Upload with automatic validation and plausibility checks
- XML Viewer with hierarchical tree navigation and node inspection
- Automatic AAS generation from KBL/VEC files via a guided wizard

These features work together to provide a seamless workflow from raw engineering data to structured AAS representations.

2. System Requirements

To use the extended BaSyx Web UI, the following requirements must be met:

- Running instance of the **BaSyx AAS Web UI** and **AAS Server environment** (More info on how to set up: https://github.com/eclipse-basyx/basyx-aas-web-ui/blob/main/README.md)
- Modern web browser (Google Chrome, Microsoft Edge, or Mozilla Firefox recommended)

3. Functionalities

This section describes the main features of the system and how to use them in practice.

3.1 File Upload with Validation

Users can upload XML-based engineering files such as **KBL, VEC, or general XML files** into the BaSyx system via the Submodel editor.

After the upload is triggered, the system automatically performs a validation process that includes:

- Checking file structure integrity
- Detecting file content type (MIME-Type detection)
- Verifying compatibility with supported XML schemas

This ensures that only valid and processable files are integrated into the Asset Administration Shell (AAS).

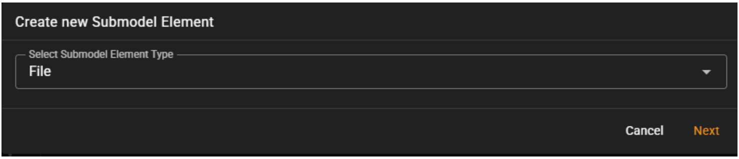
Steps

1. Open Submodel Tree

Open the desired **Submodel Tree** inside the AAS Editor, to which you want to add the new file.Then Navigate to **"Add Submodel Element"** in the context menu.

2. Select File Type

Select **Type: File**.



Create new Submodel Element

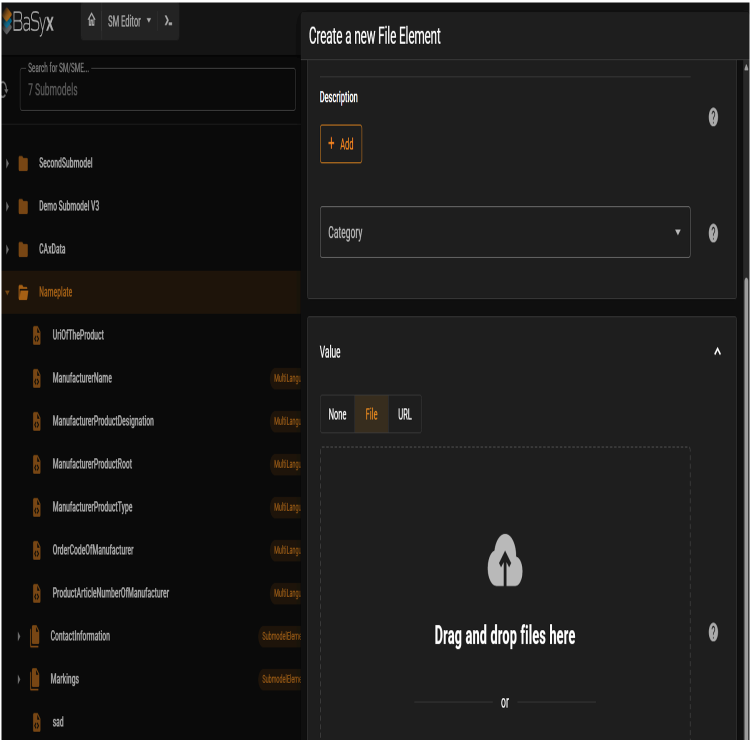
Select Submodel Element Type: File

Cancel
Next

3. Configure File Element

A new configuration window opens.

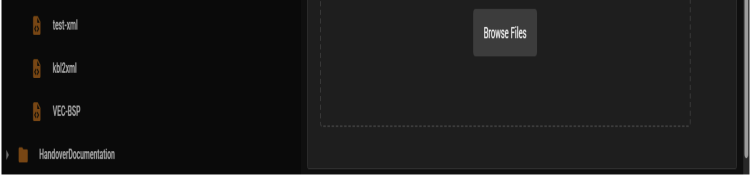
Enter the required information in the input fields and select **File** as the value type.



Create a new File Element

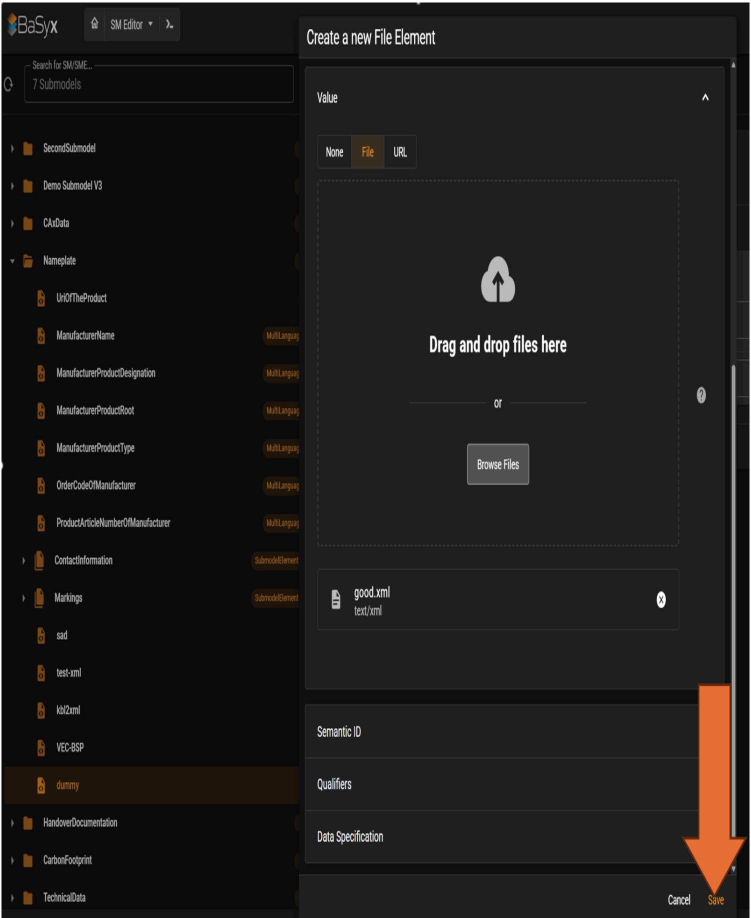
Value: File

BaSyx
SM Editor
Search for SM/SME...
7 Submodels
SecondSubmodel
Demo Submodel V3
CAxData
Nameplate
UriOfTheProduct
ManufacturerName
MultiLangu
ManufacturerProductDesignation
MultiLangu
ManufacturerProductRoot
MultiLangu
ManufacturerProductType
MultiLangu
OrderCodeOfManufacturer
MultiLangu
ProductArticleNumberOfManufacturer
MultiLangu
ContactInformation
SubmodelEleme
Markings
SubmodelEleme
sad
Description
+ Add
Category
Value
None
File
URL
Drag and drop files here
or


test-xml
kbl2xml
VEC-BSP
HandoverDocumentation
Browse Files

4. Upload File and Save

Upload your desired XML-based file and press **Save** to confirm the creation.


BaSyx
SM Editor
Search for SM/SME...
7 Submodels
SecondSubmodel
Demo Submodel V3
CAxData
Nameplate
UriOfTheProduct
ManufacturerName
MultiLanguag
ManufacturerProductDesignation
MultiLanguag
ManufacturerProductRoot
MultiLanguag
ManufacturerProductType
MultiLanguag
OrderCodeOfManufacturer
MultiLanguag
ProductArticleNumberOfManufacturer
MultiLanguag
ContactInformation
SubmodelElement
Markings
SubmodelElement
sad
test-xml
kbl2xml
VEC-BSP
dummy
HandoverDocumentation
CarbonFootprint
TechnicalData
Create a new File Element
Value
None
File
URL
Drag and drop files here
or
Browse Files
good.xml
text/xml
Semantic ID
Qualifiers
Data Specification
Cancel
Save

Result

Valid File

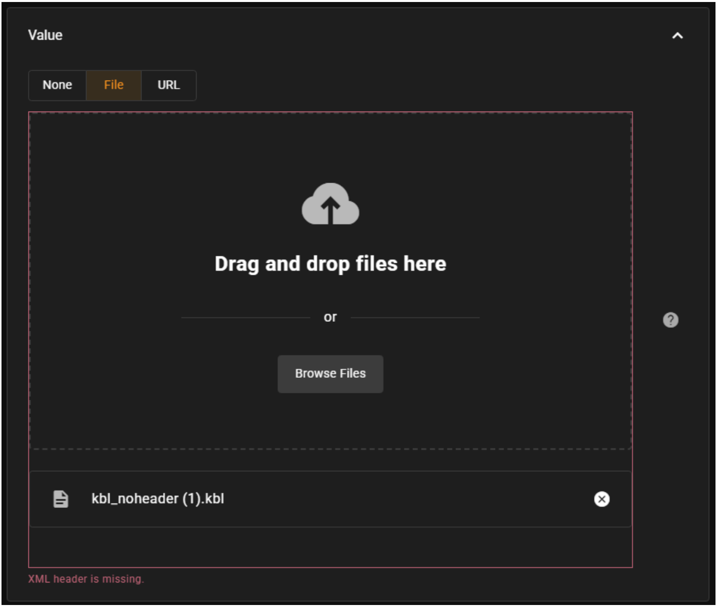
- The file is accepted successfully
- It is added to the Submodel tree
- It becomes available for further processing (e.g. XML Viewer, AAS generation)

Invalid File

If the file does not pass validation, an error message is displayed explaining the reason for rejection.

Typical error cases include:

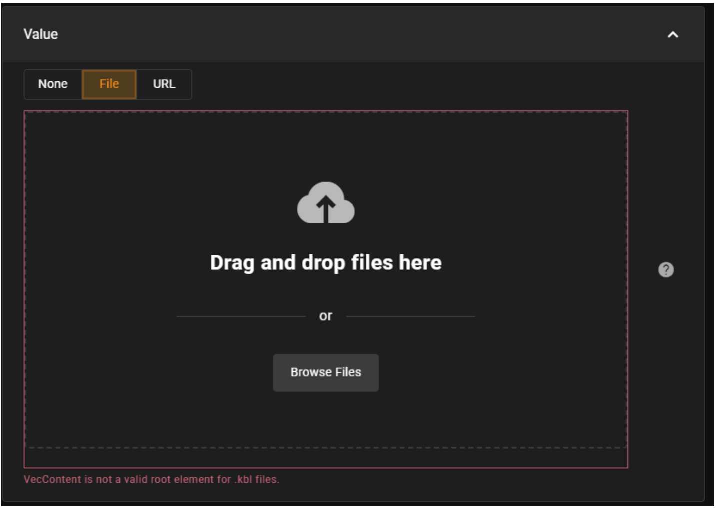
Missing XML Header


Value
None
File
URL
Drag and drop files here
or
Browse Files

kbl_noheader (1).kbl

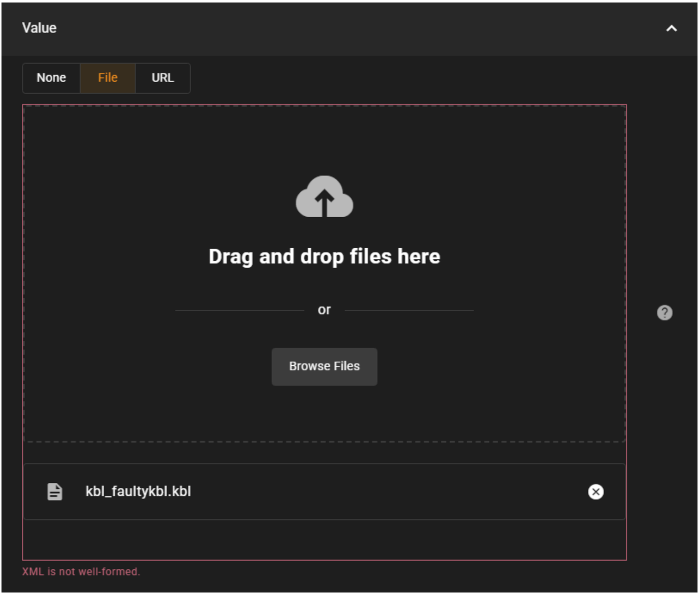
XML header is missing.

Invalid Elements / Structure Violation


Value
None
File
URL
Drag and drop files here
or
Browse Files

VecContent is not a valid root element for .kbl files.

Not Well-Formed XML File


Value
None
File
URL
Drag and drop files here
or
Browse Files
kbl_faultykbl.kbl

XML is not well-formed.

3.2 XML Viewer

Description

The XML Viewer provides a structured and interactive representation of uploaded XML-based files.

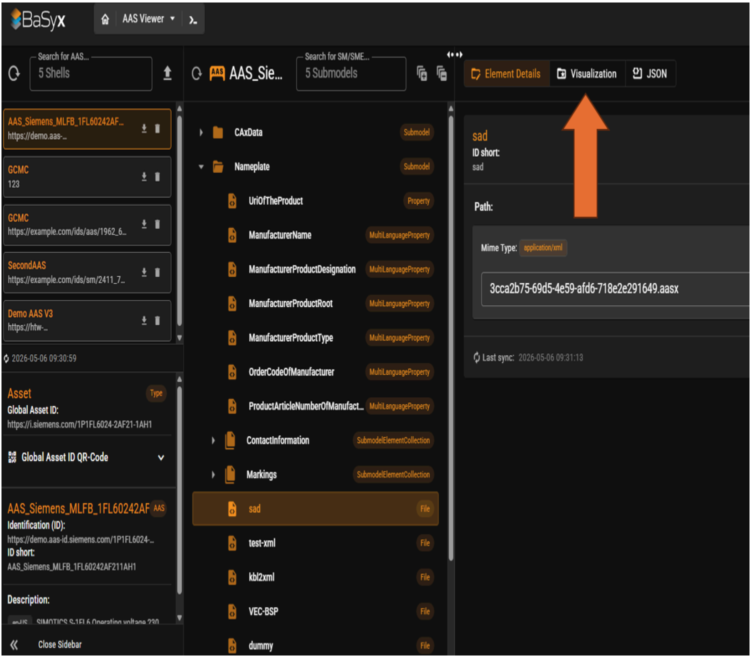
Instead of displaying raw XML text, the content is visualized as a **tree structure**, allowing users to:

- Navigate through hierarchical data
- Expand and collapse XML nodes
- Inspect attributes, values, and identifiers (IDs)
- Quickly locate relevant sections of the file

This improves readability and simplifies working with complex engineering data structures.

How to Open the XML Viewer

1. Click on an already uploaded XML-based file (e.g. XML, KBL, VEC) in the BaSyx UI
2. Open the XML Viewer by clicking the **"Visualization"** button


BaSyx
AAS Viewer
Search for AAS...
5 Shells
AAS_Siemens_MLFB_1FL60242AF...
https://demo.aas-...
GCMC
123
GCMC
https://example.com/ids/aas/1962_6...
SecondAAS
https://example.com/ids/sm/2411_7...
Demo AAS V3
https://htw-...
2026-05-06 09:30:59
Asset
Type
Global Asset ID:
https://i.siemens.com/1P1FL6024-2AF21-1AH1
Global Asset ID QR-Code
AAS_Siemens_MLFB_1FL60242AF
AAS
Identification (ID):
https://demo.aas-id.siemens.com/1P1FL6024-...
ID short:
AAS_Siemens_MLFB_1FL60242AF211AH1
Description:
Close Sidebar
AAS_Sie...
Search for SM/SME...
5 Submodels
CAxData
Submodel
Nameplate
Submodel
UriOfTheProduct
Property
ManufacturerName
MultiLanguageProperty
ManufacturerProductDesignation
MultiLanguageProperty
ManufacturerProductRoot
MultiLanguageProperty
ManufacturerProductType
MultiLanguageProperty
OrderCodeOfManufacturer
MultiLanguageProperty
ProductArticleNumberOfManufact...
MultiLanguageProperty
ContactInformation
SubmodelElementCollection
Markings
SubmodelElementCollection
sad
File
test-xml
File
kbl2xml
File
VEC-BSP
File
dummy
File
Element Details
Visualization
JSON
sad
ID short:
sad
Path:
Mime Type:
application/xml
3cca2b75-69d5-4e59-afd6-718e2e291649.aasx
Last sync:
2026-05-06 09:31:13

Layout Overview

After opening the viewer, the screen is divided into two parts:

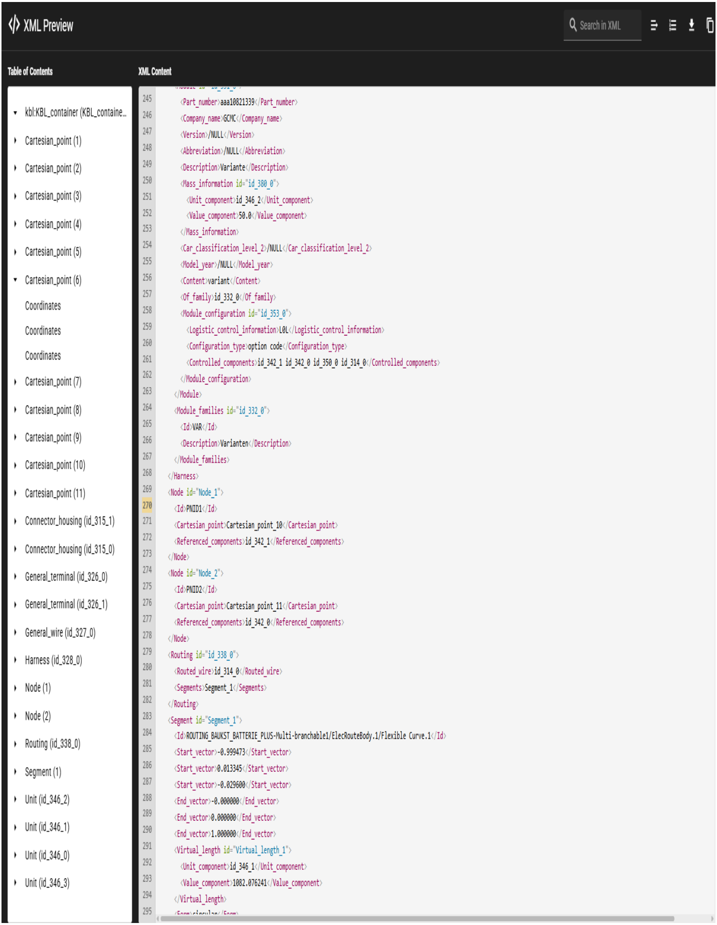
Left Side – Tree Structure

- Displays the XML document as a hierarchical tree
- Each node represents an XML element
- Elements can be expanded or collapsed

- Shows the structure of the file (parent-child relationships)

Right Side – XML Content View

- Shows the full document structure in xml form

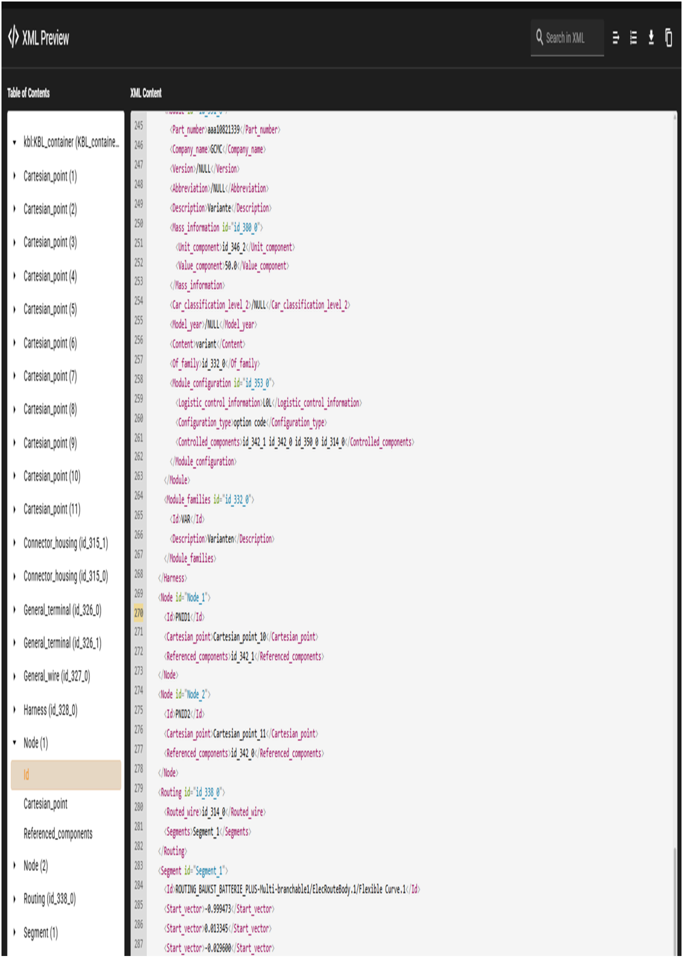

XML Preview
Search in XML
Table of Contents
XML Content
kbl:KBL_container (KBL_containe...
Cartesian_point (1)
Cartesian_point (2)
Cartesian_point (3)
Cartesian_point (4)
Cartesian_point (5)
Cartesian_point (6)
Coordinates
Coordinates
Coordinates
Cartesian_point (7)
Cartesian_point (8)
Cartesian_point (9)
Cartesian_point (10)
Cartesian_point (11)
Connector_housing (id_315_1)
Connector_housing (id_315_0)
General_terminal (id_326_0)
General_terminal (id_326_1)
General_wire (id_327_0)
Harness (id_328_0)
Node (1)
Node (2)
Routing (id_338_0)
Segment (1)
Unit (id_346_2)
Unit (id_346_1)
Unit (id_346_0)
Unit (id_346_3)

```xml
    <Part_number>aaa10821339</Part_number>
    <Company_name>GCMC</Company_name>
    <Version>/NULL</Version>
    <Abbreviation>/NULL</Abbreviation>
    <Description>Variante</Description>
    <Mass_information id="id_380_0">
      <Unit_component>id_346_2</Unit_component>
      <Value_component>50.0</Value_component>
    </Mass_information>
    <Car_classification_level_2>/NULL</Car_classification_level_2>
    <Model_year>/NULL</Model_year>
    <Content>variant</Content>
    <Of_family>id_332_0</Of_family>
    <Module_configuration id="id_353_0">
      <Logistic_control_information>L0L</Logistic_control_information>
      <Configuration_type>option code</Configuration_type>
      <Controlled_components>id_342_1 id_342_0 id_350_0 id_314_0</Controlled_components>
    </Module_configuration>
  </Module>
  <Module_families id="id_332_0">
    <Id>VAR</Id>
    <Description>Varianten</Description>
  </Module_families>
</Harness>
<Node id="Node_1">
  <Id>PNID1</Id>
  <Cartesian_point>Cartesian_point_10</Cartesian_point>
  <Referenced_components>id_342_1</Referenced_components>
</Node>
<Node id="Node_2">
  <Id>PNID2</Id>
  <Cartesian_point>Cartesian_point_11</Cartesian_point>
  <Referenced_components>id_342_0</Referenced_components>
</Node>
<Routing id="id_338_0">
  <Routed_wire>id_314_0</Routed_wire>
  <Segments>Segment_1</Segments>
</Routing>
<Segment id="Segment_1">
  <Id>ROUTING_BAUKST_BATTERIE_PLUS-Multi-branchable1/ElecRouteBody.1/Flexible Curve.1</Id>
  <Start_vector>-0.999473</Start_vector>
  <Start_vector>0.013345</Start_vector>
  <Start_vector>-0.029600</Start_vector>
  <End_vector>-0.000000</End_vector>
  <End_vector>0.000000</End_vector>
  <End_vector>1.000000</End_vector>
  <Virtual_length id="Virtual_length_1">
    <Unit_component>id_346_1</Unit_component>
    <Value_component>1082.076241</Value_component>
  </Virtual_length>
```

Navigation & Interaction

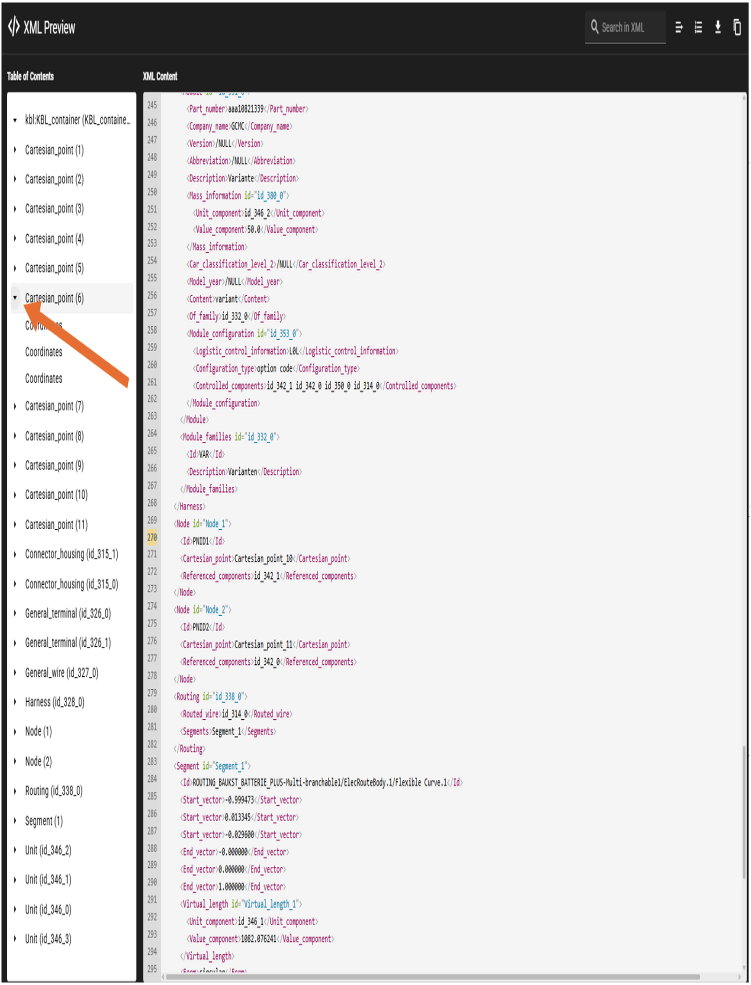
- Clicking on a node in the tree view:
  - Navigates to the corresponding XML element

- Highlights the selected element in the XML view


XML Preview
Search in XML
Table of Contents
XML Content
kbl:KBL_container (KBL_containe...
Cartesian_point (1)
Cartesian_point (2)
Cartesian_point (3)
Cartesian_point (4)
Cartesian_point (5)
Cartesian_point (6)
Cartesian_point (7)
Cartesian_point (8)
Cartesian_point (9)
Cartesian_point (10)
Cartesian_point (11)
Connector_housing (id_315_1)
Connector_housing (id_315_0)
General_terminal (id_326_0)
General_terminal (id_326_1)
General_wire (id_327_0)
Harness (id_328_0)
Node (1)
Id
Cartesian_point
Referenced_components
Node (2)
Routing (id_338_0)
Segment (1)

```xml
    <Part_number>aaa10821339</Part_number>
    <Company_name>GCMC</Company_name>
    <Version>/NULL</Version>
    <Abbreviation>/NULL</Abbreviation>
    <Description>Variante</Description>
    <Mass_information id="id_380_0">
      <Unit_component>id_346_2</Unit_component>
      <Value_component>50.0</Value_component>
    </Mass_information>
    <Car_classification_level_2>/NULL</Car_classification_level_2>
    <Model_year>/NULL</Model_year>
    <Content>variant</Content>
    <Of_family>id_332_0</Of_family>
    <Module_configuration id="id_353_0">
      <Logistic_control_information>L0L</Logistic_control_information>
      <Configuration_type>option code</Configuration_type>
      <Controlled_components>id_342_1 id_342_0 id_350_0 id_314_0</Controlled_components>
    </Module_configuration>
  </Module>
  <Module_families id="id_332_0">
    <Id>VAR</Id>
    <Description>Varianten</Description>
  </Module_families>
</Harness>
<Node id="Node_1">
  <Id>PNID1</Id>
  <Cartesian_point>Cartesian_point_10</Cartesian_point>
  <Referenced_components>id_342_1</Referenced_components>
</Node>
<Node id="Node_2">
  <Id>PNID2</Id>
  <Cartesian_point>Cartesian_point_11</Cartesian_point>
  <Referenced_components>id_342_0</Referenced_components>
</Node>
<Routing id="id_338_0">
  <Routed_wire>id_314_0</Routed_wire>
  <Segments>Segment_1</Segments>
</Routing>
<Segment id="Segment_1">
  <Id>ROUTING_BAUKST_BATTERIE_PLUS-Multi-branchable1/ElecRouteBody.1/Flexible Curve.1</Id>
  <Start_vector>-0.999473</Start_vector>
  <Start_vector>0.013345</Start_vector>
  <Start_vector>-0.029600</Start_vector>
```

- Users can expand and collapse nodes to explore the structure step by step by clicking on the arrow


XML Preview
Search in XML
Table of Contents
XML Content
kbl:KBL_container (KBL_containe...
Cartesian_point (1)
Cartesian_point (2)
Cartesian_point (3)
Cartesian_point (4)
Cartesian_point (5)
Cartesian_point (6)
Coordinates
Coordinates
Coordinates
Cartesian_point (7)
Cartesian_point (8)
Cartesian_point (9)
Cartesian_point (10)
Cartesian_point (11)
Connector_housing (id_315_1)
Connector_housing (id_315_0)
General_terminal (id_326_0)
General_terminal (id_326_1)
General_wire (id_327_0)
Harness (id_328_0)
Node (1)
Node (2)
Routing (id_338_0)
Segment (1)
Unit (id_346_2)
Unit (id_346_1)
Unit (id_346_0)
Unit (id_346_3)

```xml
      <Part_number>aaa10821339</Part_number>
      <Company_name>GCMC</Company_name>
      <Version>/NULL</Version>
      <Abbreviation>/NULL</Abbreviation>
      <Description>Variante</Description>
      <Mass_information id="id_380_0">
        <Unit_component>id_346_2</Unit_component>
        <Value_component>50.0</Value_component>
      </Mass_information>
      <Car_classification_level_2>/NULL</Car_classification_level_2>
      <Model_year>/NULL</Model_year>
      <Content>variant</Content>
      <Of_family>id_332_0</Of_family>
      <Module_configuration id="id_353_0">
        <Logistic_control_information>L0L</Logistic_control_information>
        <Configuration_type>option code</Configuration_type>
        <Controlled_components>id_342_1 id_342_0 id_350_0 id_314_0</Controlled_components>
      </Module_configuration>
    </Module>
    <Module_families id="id_332_0">
      <Id>VAR</Id>
      <Description>Varianten</Description>
    </Module_families>
  </Harness>
  <Node id="Node_1">
    <Id>PNID1</Id>
    <Cartesian_point>Cartesian_point_10</Cartesian_point>
    <Referenced_components>id_342_1</Referenced_components>
  </Node>
  <Node id="Node_2">
    <Id>PNID2</Id>
    <Cartesian_point>Cartesian_point_11</Cartesian_point>
    <Referenced_components>id_342_0</Referenced_components>
  </Node>
  <Routing id="id_338_0">
    <Routed_wire>id_314_0</Routed_wire>
    <Segments>Segment_1</Segments>
  </Routing>
  <Segment id="Segment_1">
    <Id>ROUTING_BAUKST_BATTERIE_PLUS-Multi-branchable1/ElecRouteBody.1/Flexible Curve.1</Id>
    <Start_vector>-0.999473</Start_vector>
    <Start_vector>0.013345</Start_vector>
    <Start_vector>-0.029600</Start_vector>
    <End_vector>-0.000000</End_vector>
    <End_vector>0.000000</End_vector>
    <End_vector>1.000000</End_vector>
    <Virtual_length id="Virtual_length_1">
      <Unit_component>id_346_1</Unit_component>
      <Value_component>1082.076241</Value_component>
    </Virtual_length>
    <Form>circular</Form>
```

3.3 Generate AAS from File

Description

This feature enables the automatic generation of an **Asset Administration Shell (AAS)** based on uploaded **KBL** or **VEC** files.

A guided wizard analyzes the selected file and extracts relevant engineering data. Users can review and customize the extracted information before generating the final AAS structure.

The system automatically creates:

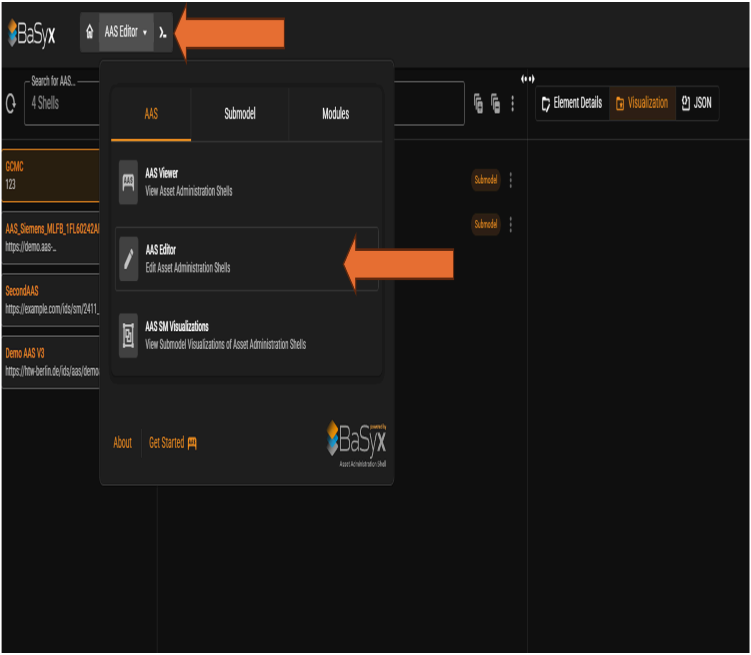
- A structured AAS instance
- Generated submodels
- Pre-filled technical data fields based on the uploaded file

---

Steps

1. Switch to Editor Mode

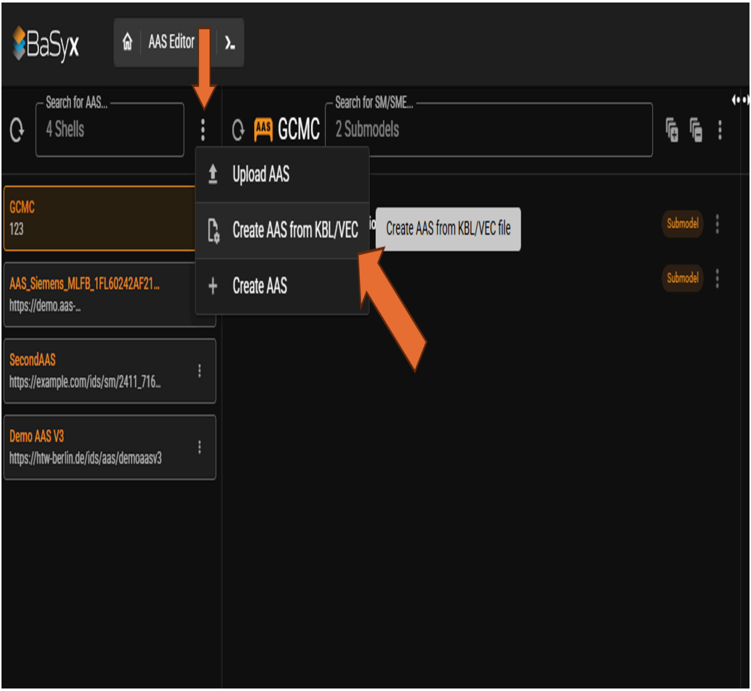
Switch to the **Editor Mode** in the BaSyx UI.


BaSyx
AAS Editor
Search for AAS...
4 Shells
AAS
Submodel
Modules
GCMC
123
AAS_Siemens_MLFB_1FL60242AI
https://demo.aas-...
SecondAAS
https://example.com/ids/sm/2411_
Demo AAS V3
https://htw-berlin.de/ids/aas/demo
AAS Viewer
View Asset Administration Shells
AAS Editor
Edit Asset Administration Shells
AAS SM Visualizations
View Submodel Visualizations of Asset Administration Shells
About
Get Started
powered by BaSyx Asset Administration Shell
Submodel
Submodel
Element Details
Visualization
JSON

---

2. Open AAS Generator

Click on the **three-dot menu (⋮)** and select **"Create AAS from KBL/VEC"**.

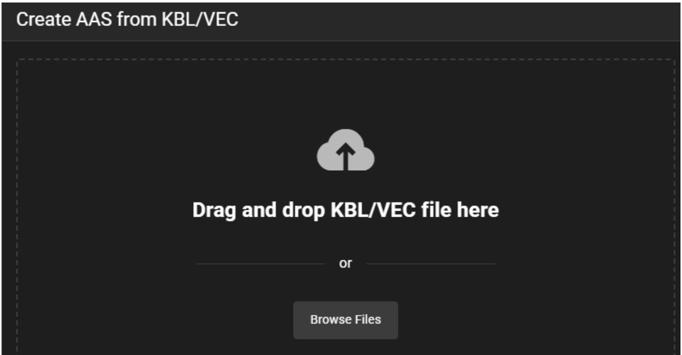

BaSyx
AAS Editor
Search for AAS...
4 Shells
Search for SM/SME...
GCMC
2 Submodels
Upload AAS
Create AAS from KBL/VEC
Create AAS
Create AAS from KBL/VEC file
GCMC
123
AAS_Siemens_MLFB_1FL60242AF21...
https://demo.aas-...
SecondAAS
https://example.com/ids/sm/2411_716...
Demo AAS V3
https://htw-berlin.de/ids/aas/demoaasv3
Submodel
Submodel

---

3. Upload Files

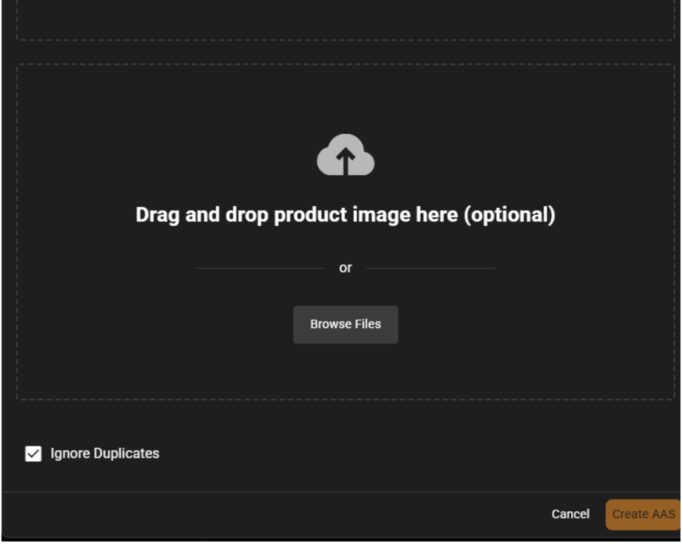
A setup dialog will appear.

- Upload a **KBL or VEC file** in the upper section (required)
- Optionally, upload a **product image** in the lower section via drag & drop or file selection


Create AAS from KBL/VEC

Drag and drop KBL/VEC file here

or
Browse Files


Drag and drop product image here (optional)

or
Browse Files
☑ Ignore Duplicates
Cancel
Create AAS

---

4. Configure AAS Metadata

After uploading the file, additional configuration fields become visible.

You can review and modify:

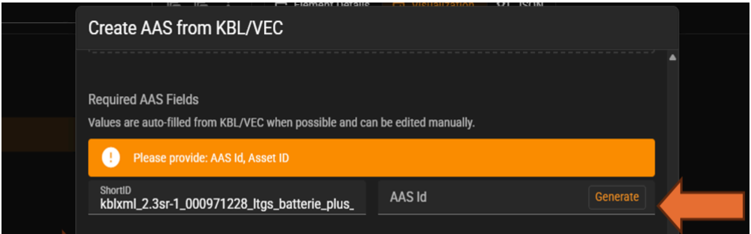
- **Short ID**
- **AAS Name**
- **Description**
- **Asset Kind**

You must generate or define:

- **Asset ID**
- **AAS ID**

These IDs can be:

- Generated automatically using the provided buttons (recommended)
- Manually defined by the user


Create AAS from KBL/VEC

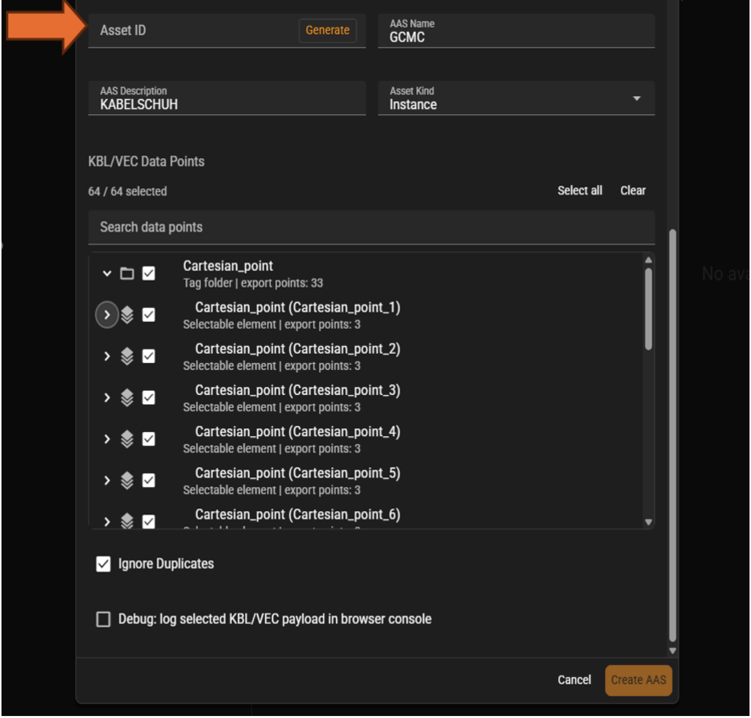
Required AAS Fields

Values are auto-filled from KBL/VEC when possible and can be edited manually.

Please provide: AAS Id, Asset ID

ShortID: kblxml_2.3sr-1_000971228_ltgs_batterie_plus_

AAS Id
Generate


Generate
Search data points
Select all
Clear
Cancel
Create AAS

Asset ID

AAS Name: GCMC

AAS Description: KABELSCHUH

Asset Kind: Instance

KBL/VEC Data Points

64 / 64 selected

- ☑ Cartesian_point
  Tag folder | export points: 33
  - ☑ Cartesian_point (Cartesian_point_1)
    Selectable element | export points: 3
  - ☑ Cartesian_point (Cartesian_point_2)
    Selectable element | export points: 3
  - ☑ Cartesian_point (Cartesian_point_3)
    Selectable element | export points: 3
  - ☑ Cartesian_point (Cartesian_point_4)
    Selectable element | export points: 3
  - ☑ Cartesian_point (Cartesian_point_5)
    Selectable element | export points: 3
  - ☑ Cartesian_point (Cartesian_point_6)

☑ Ignore Duplicates

☐ Debug: log selected KBL/VEC payload in browser console

---

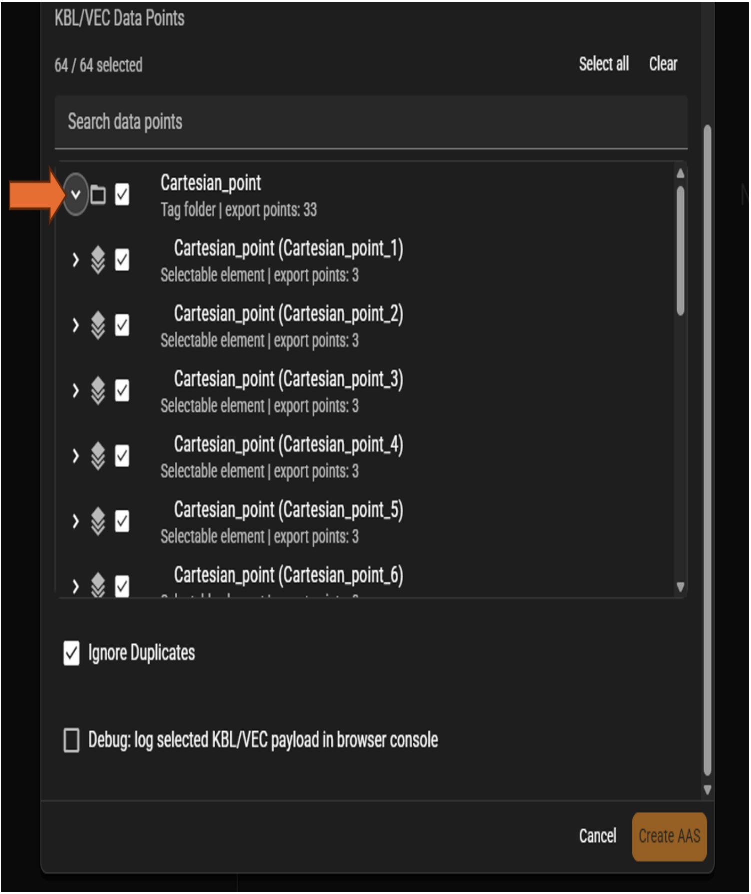
5. Review and Select Data Points

Scroll down to view the extracted data from the uploaded file.

- The data is displayed in a **tree structure**
- Each folder contains related data points
- By default, all data points are selected

You can:

- Expand folders to inspect content
- Select or deselect specific data points using checkboxes


KBL/VEC Data Points
64 / 64 selected
Select all
Clear
Search data points

☑ **Cartesian_point**
Tag folder | export points: 33

☑ Cartesian_point (Cartesian_point_1)
Selectable element | export points: 3

☑ Cartesian_point (Cartesian_point_2)
Selectable element | export points: 3

☑ Cartesian_point (Cartesian_point_3)
Selectable element | export points: 3

☑ Cartesian_point (Cartesian_point_4)
Selectable element | export points: 3

☑ Cartesian_point (Cartesian_point_5)
Selectable element | export points: 3

☑ Cartesian_point (Cartesian_point_6)

☑ Ignore Duplicates

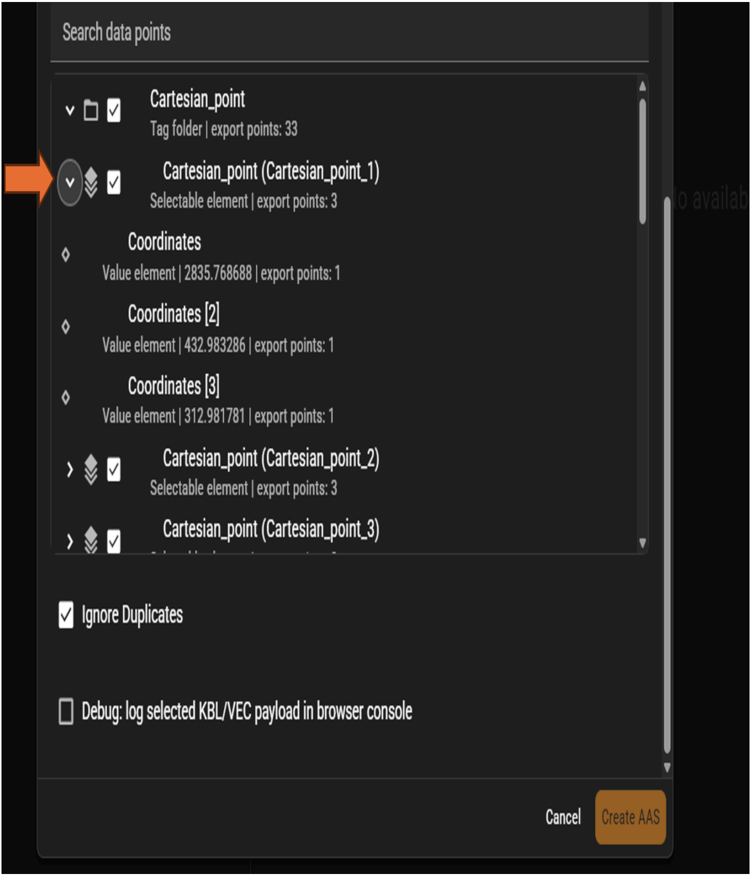
☐ Debug: log selected KBL/VEC payload in browser console

Cancel
Create AAS

---

6. Inspect Data Details (Optional)

You can open individual data points to inspect their detailed values before including them in the AAS.



Search data points

☑ Cartesian_point
Tag folder | export points: 33

☑ Cartesian_point (Cartesian_point_1)
Selectable element | export points: 3

Coordinates
Value element | 2835.768688 | export points: 1

Coordinates [2]
Value element | 432.983286 | export points: 1

Coordinates [3]
Value element | 312.981781 | export points: 1

☑ Cartesian_point (Cartesian_point_2)
Selectable element | export points: 3

☑ Cartesian_point (Cartesian_point_3)

☑ Ignore Duplicates

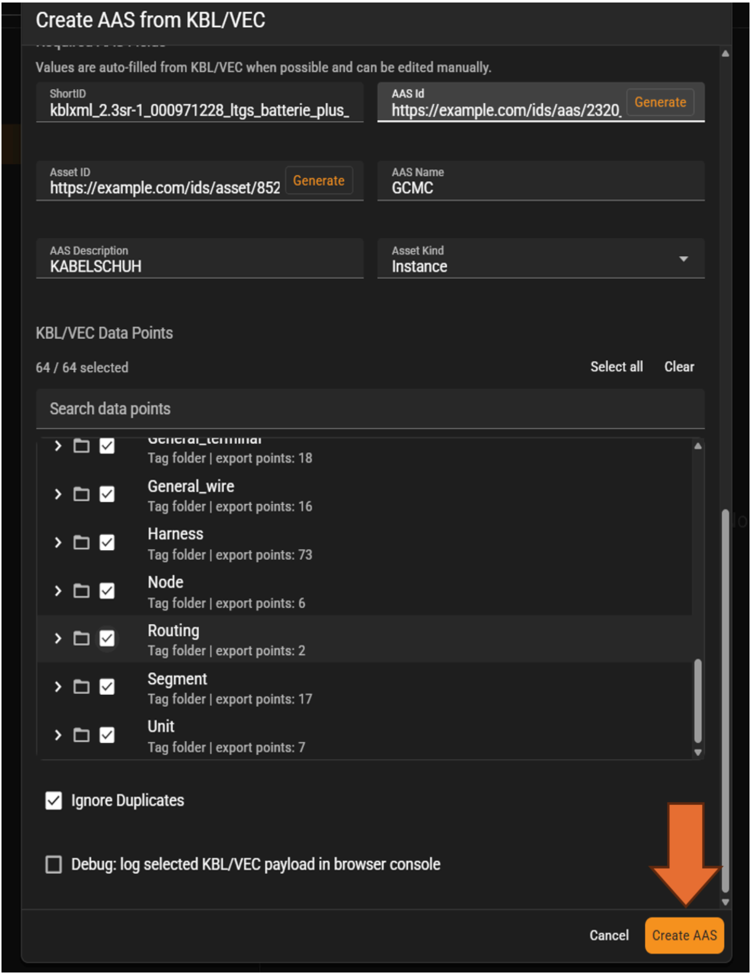
☐ Debug: log selected KBL/VEC payload in browser console

Cancel
Create AAS

---

7. Create the AAS

Finalize the process by clicking **"Create AAS"**.



Create AAS from KBL/VEC

Values are auto-filled from KBL/VEC when possible and can be edited manually.

ShortID: kblxml_2.3sr-1_000971228_ltgs_batterie_plus_

AAS Id: https://example.com/ids/aas/2320_

Generate

Asset ID: https://example.com/ids/asset/852

Generate

AAS Name: GCMC

AAS Description: KABELSCHUH

Asset Kind: Instance

KBL/VEC Data Points

64 / 64 selected

Select all
Clear
Search data points

- ☑ General_terminal — Tag folder | export points: 18
- ☑ General_wire — Tag folder | export points: 16
- ☑ Harness — Tag folder | export points: 73
- ☑ Node — Tag folder | export points: 6
- ☑ Routing — Tag folder | export points: 2
- ☑ Segment — Tag folder | export points: 17
- ☑ Unit — Tag folder | export points: 7

☑ Ignore Duplicates

☐ Debug: log selected KBL/VEC payload in browser console

Cancel
Create AAS

---

8. Completion

The AAS is generated and uploaded to the system.

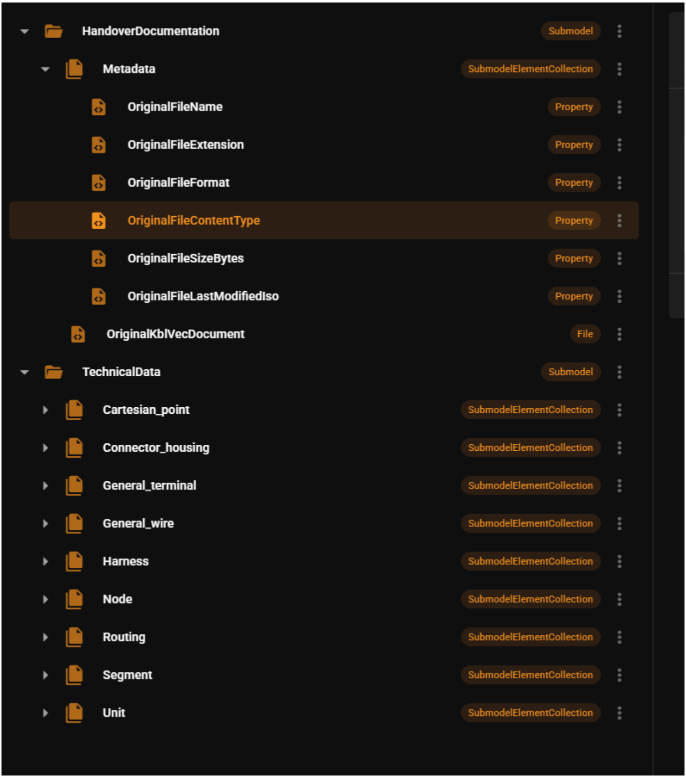
Result

A new AAS is created containing at least the following submodels:

- **HandoverDocumentation**

Contains the original uploaded file, including metadata such as file name, format, and reference.

- **TechnicalData**
  Contains the extracted and structured engineering data selected by the user.



HandoverDocumentation — Submodel
- Metadata — SubmodelElementCollection
  - OriginalFileName — Property
  - OriginalFileExtension — Property
  - OriginalFileFormat — Property
  - OriginalFileContentType — Property
  - OriginalFileSizeBytes — Property
  - OriginalFileLastModifiedIso — Property
- OriginalKblVecDocument — File

TechnicalData — Submodel
- Cartesian_point — SubmodelElementCollection
- Connector_housing — SubmodelElementCollection
- General_terminal — SubmodelElementCollection
- General_wire — SubmodelElementCollection
- Harness — SubmodelElementCollection
- Node — SubmodelElementCollection
- Routing — SubmodelElementCollection
- Segment — SubmodelElementCollection
- Unit — SubmodelElementCollection

5. Copyright / License

License

This project is based on the Eclipse BaSyx AAS Web UI, which is licensed under the MIT License.

All additional functionalities, plugins, and extensions developed within this project are:

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